

Task 2

1. Find the total sales, average sale value, and total quantity sold.

```
mysql> use retail_sales;
Database changed
mysql> SELECT
  -> SUM(o.quantity * p.price) AS total_sales,
  -> AVG(o.quantity * p.price) AS avg_sale_value,
  -> SUM(o.quantity) AS total_quantity_sold
  -> FROM orders o
  -> JOIN products p ON o.product_id = p.product_id;
+-----+-----+-----+
| total_sales | avg_sale_value | total_quantity_sold |
+-----+-----+-----+
| 914305.96 | 436.839924 | 6199 |
+-----+-----+-----+
1 row in set (0.01 sec)
```

2. Find the total sales by product category and identify the highest-selling category.

```
mysql> SELECT
  -> p.category,
  -> SUM(o.quantity * p.price) AS total_sales
  -> FROM orders o
  -> JOIN products p ON o.product_id = p.product_id
  -> GROUP BY p.category
  -> ORDER BY total_sales DESC;
+-----+-----+
| category | total_sales |
+-----+-----+
| Home | 272478.14 |
| Clothing | 199263.54 |
| Sports | 193083.93 |
| Beauty | 161715.34 |
| Electronics | 87765.01 |
+-----+-----+
5 rows in set (0.01 sec)
```

3. Find the top 5 customers by total spending.

```
mysql> SELECT
  -> o.customer_id,
  -> SUM(o.quantity * p.price) AS total_spent
  -> FROM orders o
  -> JOIN products p ON o.product_id = p.product_id
  -> GROUP BY o.customer_id
  -> ORDER BY total_spent DESC
  -> LIMIT 5;
+-----+-----+
| customer_id | total_spent |
+-----+-----+
| C0299 | 3718.95 |
| C0910 | 3710.21 |
| C0334 | 3682.70 |
| C0335 | 3682.38 |
| C0588 | 3660.70 |
+-----+-----+
5 rows in set (0.01 sec)
```

4. Find the number of orders and total sales for each customer using a JOIN.

```
mysql> SELECT
->   c.customer_id,
->   COUNT(o.order_id) AS num_orders,
->   SUM(o.quantity * p.price) AS total_sales
-> FROM customers c
-> JOIN orders o ON c.customer_id = o.customer_id
-> JOIN products p ON o.product_id = p.product_id
-> GROUP BY c.customer_id;
```

customer_id	num_orders	total_sales
C0692	5	1019.47
C1187	4	889.29
C0290	3	1503.95
C0416	1	124.32
C0824	2	403.70
C0248	3	933.48
C1192	4	2194.02
C0771	4	586.24
C0014	3	1149.00
C0582	2	859.20
C0181	2	404.19
C0450	1	106.34
C0972	4	1740.94
C0089	2	1126.92
C0929	5	2359.35
C0408	2	891.38
C0467	4	1535.78
C0410	4	1089.92
C1199	3	271.12
C0443	6	3066.52
C0735	2	920.96
C0066	3	940.49
C0228	2	1056.73
C0844	1	426.80
C0970	3	399.56
C1175	2	1266.78
C0493	2	1528.22
C0625	3	1735.05
C1196	2	1156.20
C0284	2	770.52
C0951	2	317.04
C0553	4	1825.75
C0425	3	2371.96
C0113	2	528.63
C0138	6	2300.38
C0489	1	253.94
C1189	1	413.52
C0130	1	632.22
C1090	2	615.26
C0754	3	594.38
C0165	2	257.22
C0158	1	171.52
C0304	3	878.46
C0047	3	2395.16
C0271	4	1980.09
C0044	3	577.77
C0476	3	510.43
C0703	4	1564.15
C1162	3	1467.61
C0533	2	1176.07
C0591	2	930.80
C0315	4	1187.17
C0758	3	918.23

C0734	1	217.47
C1137	2	1517.08
C0471	1	1428.20
C1089	1	1033.80
C0524	1	660.16
C0986	1	101.16
C1107	2	331.43
C0240	1	264.18
C0005	1	58.58
C0500	1	1293.90
C0201	1	27.81
C0865	1	206.54
C0805	1	632.22
C0941	1	660.45
C0994	1	289.84
C1046	1	36.66
C0936	1	40.13
C0283	1	33.21
C0379	1	48.44
C0860	1	193.76
C0720	1	201.61
C0671	1	80.26
C1163	2	987.77
C0892	1	111.24
C0634	1	1104.10
C0760	1	828.69
C0277	1	480.56
C0357	1	761.82
C0793	2	1082.82
C0802	1	360.42
C1108	1	806.44
C0706	1	392.52
C0193	1	284.85
C1100	2	522.04
C0571	1	33.21
C0426	1	733.74
C1030	1	56.97
C0575	1	182.71
C0586	1	733.74
C0038	1	1104.10
C0096	1	190.84
C0617	1	1381.15
C0178	1	272.92
C0907	1	1381.15
C0867	1	41.44
C0520	1	213.45
C1008	1	168.81
C0380	1	730.84
C0723	1	73.32
C0422	1	733.74
C0887	1	294.52
C0458	1	269.97
C0729	1	171.52
C0966	1	1363.80
C0454	1	258.78
C0420	1	80.26
C0221	1	175.74
C0653	1	186.36
C0693	1	480.56
C0960	1	217.47
C0759	1	206.76
C0355	1	1381.15

+-----+-----+-----+

980 rows in set (0.01 sec)